Content

1.
2.
3.
4.
5.
6.
7.
8.
9.

Three bit supersymmetry theory

0	0	0	Particle 1	Symmetrical
1	1	1	Antiparticle 1	Symmetrical
0	1	0	Particle 2	Symmetrical
1	0	1	Antiparticle 2	Symmetrical
1	0	0	Particle 3	Asymmetrical
0	1	1	Antiparticle 3	Asymmetrical
0	0	1	Particle 4	Asymmetrical
1	1	0	Antiparticle 4	Asymmetrical

Supersymmetry in physics

Supersymmetry, or Fermi-Bose symmetry, is a hypothetical symmetry connecting bosons and fermions in nature. The abstract supersymmetry transformation links bosonic and fermionic quantum fields so that they can transform into each other. Figuratively, we can say that the supersymmetry transformation can transform matter into interaction (or into radiation), and vice versa.

Supersymmetry involves doubling (at least) the number of known elementary particles due to the presence of superpartners. For a photon - photino, quark - squark, Higgs - higgsino, W-boson - wine, gluon - gluino, and so on. Superpartners must have a spin value that is half an integer different from the spin value of the original particle

A language for describing interactions using one-, two-, three-bit operations

N|P			One- and two-bit negation (Not), 0 and 1 – control bits, P – bit remains unchanged (Pass)
N|0|1|P	N|0|1|P		
S	S	P	Unconditional exchange of two bits (Swap), the third bit of P remains unchanged (Pass)
S	P	S	
P	S	S	
0|1	S	S	A control bit is supplied to one of the inputs; if the bit is zero or one, then the other two, designated as S, are swapped (Control Swap and Anti-Control Swap)
S	0|1	S	
S	S	0|1	
P	P	P	Does not perform any actions, incoming bits are sent to the output without changes (Pass)
N|P	N|P	N|P	The unconditional negation of one, two or three bits (Not)
0|1	N|P	N|P	Conditional negation (Not depending on the value of one or two control bits). A vertical bar indicates alternative values, and a P means the bit does not change (Pass).
N|P	0|1	N|P	
N|P	N|P	0|1	
0|1	0|1	N	
0|1	N	0|1	
N	0|1	0|1	

IT IS IMPORTANT. The language is not limited to three bits to describe operations; in fact, the structure of the operation can be divided into two parts: conditional control passive (0|1) and active (N|S|...)

Example No. 1. Controlled exchange and unconditional permutation

The input has three bits, each can have the value 0 or 1		The values here 0, 1 are control ones, and S values on the same vertical are swapped if the control bit is 0 or 1							The output is three bits, each can have the value 0 or 1
{0, 1}	→	0	1	S	S	S	S	→	{0, 1}
{0, 1}	→	S	S	0	1	S	S	→	{0, 1}
{0, 1}	→	S	S	S	S	0	1	→	{0, 1}

The transformation above is equivalent to swapping the first and last inputs

x1	→	x3
x2	→	x2
x3	→	x1

It is obvious that these transformations are reversible, since in the first case we used a reversible logic gate Control Swap and its opposite Anti-Control Swap, in the second case the exchange of the first and last inputs is unconditional, that is, it is also reversible.

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N3Lang is an experimental language based on three-bit supersymmetry; basic language operations have three inputs and three outputs. A program consists of a finite sequence of such instructions.

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Project N3Lang on Github: https://github.com/xayam/N3Lang